

RIT*: Riemannian Informed Trees for Cost-Adaptive Optimal Motion Planning

Abstract—We present Riemannian Informed Trees (RIT*), a planning framework that replaces Euclidean primitives in batch-informed search with their Riemannian counterparts. RIT* constructs a tighter, cost-consistent informed set, performs a nearest-neighbour search under an anisotropic distance metric, and evaluates edge costs efficiently via a cascading scheme. We further introduce a Collision-Adaptive Metric Refinement (CARM), which learns an obstacle-proximity cost field online from collision feedback, reducing the reliance on prior metric design in practical settings. Experiments across environments from 2-D to 14-D show that RIT* is competitive in low-dimensional and spatially constant-metric settings and produces substantially lower-cost solutions when the metric varies spatially in high-dimensional configuration spaces. Performance gains scale with anisotropy and dimension, reaching up to 13.0% improvement over BIT* in the 3-D anisotropic benchmark, up to 24.0% in 6-DOF manipulation, and 24.8-63.5% in a 14-DOF bimanual planning problem, where Euclidean-informed baselines degrade.

Index Terms—Motion planning, Riemannian geometry, asymptotic optimality, informed sampling, online metric learning.

I. INTRODUCTION

Sampling-based motion planning is a fundamental technique in robotics, which enables collision-free path computation in high-dimensional configuration spaces [1]. The development of asymptotically optimal (AO) planners, including Rapidly-exploring Random Trees* (RRT*) [2] and Probabilistic Roadmap Method* (PRM*), demonstrated that random sampling methods can asymptotically converge to an optimal path. Later work improved convergence by employing *informed sampling*: after an initial solution with some cost is obtained, all subsequent samples are then taken solely from the *informed set*, which corresponds to a prolate hyperspheroid in Euclidean space [3]. This idea is the basis for Informed RRT* [3], BIT* [4], AIT*, EIT* [5], and APT* [6].

These planners show promising performance; however, they select this set using Euclidean distance, which does not adequately reflect the true cost structure in anisotropic environments. In addition, existing approaches typically rely on a predetermined cost structure rather than learning it from planning experience, which reduces their effectiveness when the cost landscape is unknown or varies with the environment.

In this work, we address both limitations with RIT*. We replace Euclidean components with a Riemannian formulation, where a spatially varying metric tensor $G(x)$ captures the underlying cost structure. We further introduce a Collision-Adaptive Metric Refinement (CARM), which learns a collision-induced conformal cost field while planning from collision feedback and adapts the planner accordingly.

Contributions: The key contributions are:

- 1) *Riemannian informed planning*: RIT* replaces Euclidean components with Riemannian ones. The in-

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Fig. 1: Overview of RIT*. (a) BIT* samples from the Euclidean informed set I_E , which ignores anisotropic cost structure. (b) RIT* uses a cost-aligned Riemannian informed set aligned with the underlying cost structure, while the CARM module (red shading) learns a cost field online from collision feedback, guiding paths away from obstacles without prior knowledge.

formed set better aligns with the underlying cost structure, nearest neighbours use an anisotropic metric, and edge costs are computed via a cascading scheme to reduce computation. These changes focus sampling on relevant regions and improve efficiency.

- 2) *Online proximity-aware metric refinement via CARM*: We introduce CARM, which learns an obstacle-proximity cost field from collision feedback and applies it as a conformal scaling of a base metric. It requires no prior obstacle knowledge: given a base metric (e.g., joint-space inertia), CARM refines it with proximity-aware structure; without one, it defaults to Euclidean and learns from scratch.
- 3) *Experimental validation*: We evaluate RIT* from 2-D and 3-D to high-dimensional tasks (6-DOF, 14-DOF), including a real UR10e arm, and provide a comprehensive benchmark against asymptotically optimal planners.

II. RELATED WORK

We review prior work on informed sampling-based planning, the use of Riemannian geometry in motion planning, and obstacle-aware cost design.

Euclidean informed sampling planners: Sampling-based planners have been improved by restricting search to an informed subset of the space. Gammell et al. [3] introduced direct sampling of a prolate hyperspheroid to accelerate RRT*. BIT* [4] combines batch sampling with informed sets and lazy edge evaluation, while AIT* and EIT* [5] extend this with asymmetric bidirectional search. APT* [6] shapes nearest-neighbour search using virtual forces and adapts batch size, and FIT* [7] improves early convergence through batch adaptation. Despite these advances, all define the informed set, nearest-neighbour search, and edge cost in *Euclidean* distance.

Riemannian geometry in motion planning. Riemannian geometry naturally models anisotropic costs, but its use is largely limited to local or optimisation methods. CHOMP [8] and STOMP [9] optimise trajectories under such costs, but are not sampling-based and lack AO guarantees. In sampling-based planning, Kim et al. [10] use Riemannian metrics for local steering, and Kyaw and Kelly [11] propose a Riemannian RRT* using retractions and natural gradients. However, these methods are incremental, do not construct informed sets, and do not address global sampling. Bi-AM-RRT* [12] similarly biases nearest-neighbour selection via a diffusion-based metric, but does not integrate informed sampling or batch processing.

Obstacle-aware Riemannian metrics. Several works incorporate obstacle information into metric design but typically rely on prior knowledge or demonstrations. Klein et al. [13] design barrier-based metrics requiring full obstacle geometry, while RMPflow [14] composes task-space metrics for reactive control. Beik-Mohammadi et al. [15] learn Riemannian manifolds from demonstrations for reactive motion generation. These approaches are not integrated with sampling-based planning and require information beyond collision checking.

Key distinction: Prior Riemannian approaches use the metric primarily for *local* operations, such as steering, trajectory optimisation, or reactive control. In contrast, RIT* integrates Riemannian geometry into the *global* informed sampling pipeline. It replaces the informed set, nearest-neighbour search, edge cost evaluation, and rewiring operations of a batch-informed planner with their Riemannian counterparts. Additionally, RIT* incorporates CARM, which learns a conformal scaling online from collision feedback, enabling adaptive and cost-aware planning without prior environment knowledge.

III. PROBLEM FORMULATION

We consider optimal motion planning in a configuration space where the cost of motion is anisotropic and may vary across the space. Such costs are naturally described by a Riemannian metric, which assigns a direction-dependent traversal cost at each configuration.

Let $\mathcal{C} \subseteq \mathbb{R}^d$ denote a d -dimensional configuration space, and let $\mathcal{C}_{\text{free}} \subseteq \mathcal{C}$ be the subset of collision-free configurations. The space is equipped with a smooth, positive-definite metric tensor field $G : \mathcal{C} \rightarrow \mathbb{S}_{++}^d$, where $\mathbb{S}_{++}^d$ denotes the set of symmetric positive-definite matrices. For any configuration $x \in \mathcal{C}$, the matrix $G(x)$ defines the local cost of moving

through x , so that different directions can have different traversal costs.

Under this metric, the cost of a path $\sigma : [0, 1] \rightarrow \mathcal{C}$ is its Riemannian arc length,

$$c(\sigma) = \int_0^1 \sqrt{\dot{\sigma}(t)^\top G(\sigma(t)) \dot{\sigma}(t)} dt, \quad (1)$$

where $\dot{\sigma}(t)$ is the tangent vector of the path. The resulting *geodesic distance* $d_R(x, y)$ is defined as the minimal cost of any path between the two configurations x and y , i.e., $d_R(x, y) = \inf_{\sigma} c(\sigma)$ taken over all paths σ that connect x and y . In the special case where $G(x) = I_d$, this corresponds to the usual Euclidean distance.

Given a start state $x_s \in \mathcal{C}_{\text{free}}$ and a goal state $x_g \in \mathcal{C}_{\text{free}}$, our objective is to find a collision-free path of minimum cost:

$$\sigma^* = \arg \min_{\sigma : x_s \rightarrow x_g, \sigma \subset \mathcal{C}_{\text{free}}} c(\sigma), \quad (2)$$

where σ is any feasible path that remains in the free space. A planner is *asymptotically optimal* if the cost of its returned solution, denoted by c_n , converges almost surely to the optimal cost c^* as the number of samples $n \rightarrow \infty$.

After a feasible solution with cost c_{best} is found, the search can be restricted to configurations that may still generate a better path. In Euclidean informed planning, this region is the *informed set*

$$I_E = \{x \in \mathcal{C} : \|x_s - x\|_2 + \|x - x_g\|_2 \leq c_{\text{best}}\}, \quad (3)$$

which contains all points that could lie on a path shorter than the current best solution under Euclidean distance.

For anisotropic costs, however, Euclidean distance does not accurately reflect the true cost-to-come and cost-to-go. We therefore define the *Riemannian informed set* as

$$I_R = \{x \in \mathcal{C} : d_R(x_s, x) + d_R(x, x_g) \leq c_{\text{best}}\}. \quad (4)$$

This set contains configurations that can still improve the current solution under the Riemannian cost. The relationship between I_R and I_E depends on the metric spectrum. When $G \succeq I_d$, $d_R(x, y) \geq \|x - y\|_2$, resulting in $I_R \subseteq I_E$. More generally, under the bi-Lipschitz bounds of Sec. IV-F, the two informed sets are equivalent up to constant-factor distortion, while I_R remains aligned with the underlying cost structure.

The goal of this work is to design an AO informed planner that operates natively under the Riemannian cost: sampling from I_R , selecting neighbours and evaluating edges according to the geodesic distance d_R , and, when no base metric is available or when the available metric does not encode obstacle proximity, refining the metric online from collision feedback gathered during planning.

IV. METHOD

Batch Informed Trees [4] is an asymptotically optimal planner that combines sampling-based planning with graph search. It incrementally builds a tree using batches of samples drawn from an informed set, focusing the search on regions that can improve the current solution. Within each batch, vertices and edges are ordered by cost-to-come and heuristic estimates, and

Algorithm 1: RIT* planner

Input: $x_s, x_g, \mathcal{C}_{\text{free}}, G(\cdot), B, T, K_c, \sigma, \alpha$
Output: Optimal path σ^*

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1  $\mathcal{V} \leftarrow \{x_s\}; \mathcal{E} \leftarrow \emptyset; c_{\text{best}} \leftarrow \infty; \mathcal{C}_{\text{col}} \leftarrow \emptyset;$ 
2 for  $t = 1, \dots, T$  do
3    $X_{\text{new}} \leftarrow \text{RIEMANNIANINFSAMPLE}(B, c_{\text{best}}, G)$ 
4    $\mathcal{V} \leftarrow \mathcal{V} \cup X_{\text{new}}$ 
5    $\text{PRUNENODES}(\mathcal{V}, c_{\text{best}}, G)$ 
6   Compute  $r_n^R$  via (6)
7   foreach  $v \in \mathcal{V}$  in order of  $f(v)$  do
8      $N(v) \leftarrow \text{neighbours of } v \text{ within } r_n^R$ 
9     foreach  $u \in N(v)$  do
10       $(c_{uv}, p_{\text{col}}) \leftarrow \text{CASCADINGEDGE}(u, v, G)$ 
11       $\text{UPDATEANDREWIRE}(\mathcal{V}, \mathcal{E}, u, v, c_{uv}, r_n^R, G)$  if  $v = x_g$ 
12        and  $g(v) < c_{\text{best}}$  then
13           $c_{\text{best}} \leftarrow g(v)$ 
14        end
15        if  $p_{\text{col}} \neq \emptyset$  then
16           $\mathcal{C}_{\text{col}} \leftarrow \mathcal{C}_{\text{col}} \cup \{p_{\text{col}}\}$ 
17        end
18      end
19    end
20    if  $t \bmod K_c = 0$  and  $|\mathcal{C}_{\text{col}}| > 0$  then
21       $G \leftarrow \text{CARM\_UPDATE}(\mathcal{C}_{\text{col}}, G, \sigma, \alpha)$ 
22    end
23 end
24 return  $\sigma^*$ 

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edges are evaluated lazily to reduce computation, balancing global exploration with goal-directed search.

RIT* extends this framework with three modifications and a learning component. i) The informed set is defined using a Riemannian metric, producing a cost-aligned sampling region. ii) Nearest-neighbour search uses Riemannian distance, enabling connections along low-cost directions. iii) Edge costs are approximated via a cascading quadrature scheme. In addition, RIT* introduces CARM, which learns a conformal scaling online from collision feedback.

Algorithm 1 summarises the procedure. The planner takes as input a start-goal pair (x_s, x_g) , free space $\mathcal{C}_{\text{free}}$, a metric tensor G (or I_d if unknown), batch size B , iteration budget T , and CARM parameters: update interval K_c , bandwidth σ , and inflation strength α . A metric cache $\mathcal{M}$ is precomputed on a grid for $O(1)$ metric lookups. Within each batch, vertices are processed in order of $f(v) = g(v) + \hat{h}_R(v)$ (line 8), where $g(v)$ is the Riemannian cost-to-come and $\hat{h}_R(v)$ is an admissible heuristic: for constant metrics it equals the midpoint (L1) estimate $\bar{c}(v, x_g)$; for spatially-varying metrics it uses the guaranteed lower bound $\hat{h}_R(v) = \sqrt{\lambda_{\min}} \|v - x_g\|_2$ (Remark 3). Components Riemannian sampling (line 3), cascading evaluation (line 10), rewiring, pruning (lines 5, 10), and CARM updates (line 20) are detailed later.

A. Riemannian Informed Sampling

RIT* draws samples from the Riemannian informed set I_R (line 3 of algorithm 1), defined in (4). When $G \succeq I_d$, the Riemannian distance dominates the Euclidean one, $d_R(x, y) \geq \|x - y\|_2$, so $I_R \subseteq I_E$.

To draw samples from I_R efficiently, we apply a metric whitening transform. Let $\bar{G}$ denote the mean of G at $N_{\text{seg}} = 10$ uniformly spaced points along the start-goal segment, and $L = \text{chol}(\bar{G})$ its Cholesky factor satisfying $\bar{G} = LL^\top$. The change of coordinates $\tilde{x} = L^\top x$ maps I_R to an approximate prolate hyperspheroid, where the direct sampling of [3] applies. Samples are mapped back via $x = L^{-\top} \tilde{x}$.

figures/fig_neighbour.png

Fig. 2: Riemannian nearest-neighbour selection and cascading edge evaluation in RIT*. (a) The Riemannian ball expands along low-cost directions compared to the Euclidean ball. (b-d) Cascading evaluation filters edges with increasing accuracy, rejecting most early and evaluating only promising ones at full precision.

This transformation approximates the Riemannian-informed set; its accuracy depends on the smoothness of G and is most effective when metric variation along the start-goal path is moderate. Empirically, the median relative volume mismatch $|\text{vol}(L^{-\top} \mathcal{E}) - \text{vol}(I_R^{\text{true}})| / \text{vol}(I_R^{\text{true}})$ is below 1% on our 3-D, 6-D, and 14-D benchmarks (numerical integration error; analytically exact for the spatially-uniform diagonal metrics used there); mild spatial variations in the metrics lead to discrepancies below 3% in our experiments. To validate this under controlled spatial variation, we tested a synthetic 3-D metric $G(x) = \text{diag}(1, 3, 6)(1 + \beta \|x - x_c\|^2)$ with $\beta \in \{0, 0.5, 1, 2\}$: median mismatch grows from $< 0.5\%$ ($\beta = 0$) to 2.7% ($\beta = 1$) and 6.1% ($\beta = 2$), i.e. the whitening degrades smoothly with $\|\nabla G\|$.

B. Riemannian Nearest-Neighbour Search

Let r_n^R be the neighbour connection radius for n samples; RIT* uses the Riemannian ball $\{u : d_R(u, v) \leq r_n^R\}$, which becomes an ellipsoid aligned with the local metric. It expands along low-cost directions and contracts along high-cost directions, producing neighbour sets that better match the underlying traversal cost structure.

The corresponding connection radius is

$$r_n^R = \gamma_R \left(\frac{\log n}{n} \right)^{1/d}, \quad \gamma_R = 2 \left(1 + \frac{1}{d} \right)^{1/d} \left(\frac{\mu_R(I_R)}{\zeta_d} \right)^{1/d}, \quad (5)$$

where $\mu_R(I_R)$ is the Riemannian volume of I_R and ζ_d is the volume of the unit d -ball. Because $\mu_R(I_R) < \mu(I_E)$, this radius

is typically smaller than its Euclidean counterpart, reducing the number of candidate edges.

C. Cascading Edge Evaluation

Evaluating the Riemannian edge cost requires numerical integration of (1), involving multiple metric tensor evaluations. Since most candidate edges cannot improve the current solution, applying full-precision quadrature to all of them is computationally expensive. RIT* instead screens edges through a three-level cascade of increasing accuracy and cost, rejecting unpromising edges early (line 10 of algorithm 1).

For a candidate edge from parent u to vertex v , let $\Delta = v - u$ and $m = (u+v)/2$. An edge is discarded whenever $g(u) + \hat{c}(u, v) > c_{\text{best}}$, where $g(u)$ is the cost-to-come from x_s to u along the current tree.

L1: Midpoint check: A single metric lookup at the midpoint gives

$$\hat{c}_1 = \sqrt{\Delta^\top G(m) \Delta}. \quad (6)$$

L_1 rejects the majority of candidates across our 2-D to 14-D benchmarks (78-85%; 80% in the 6-D UR10 environment).

L2: Simpson estimate: Surviving edges are re-evaluated with Simpson's rule:

$$\hat{c}_2 = \frac{1}{6} \left(\sqrt{\Delta^\top G(u) \Delta} + 4\sqrt{\Delta^\top G(m) \Delta} + \sqrt{\Delta^\top G(v) \Delta} \right), \quad (7)$$

which captures metric variation along the edge more accurately. This level rejects approximately 75% of the edges that passed L_1 .

L3: Full quadrature and collision check: Only the roughly 5% of edges surviving both levels receive 10-point Gauss-Legendre integration followed by collision checking. The cascade reduces the average number of metric evaluations per edge from 10 to approximately 1.6.

Fig. 2 illustrates the cascade: of seven candidate neighbours, two are rejected at L_1 , one at L_2 , and one at L_3 due to collision, leaving three accepted edges.

Remark 1. Straight-line cost approximation and admissible heuristic: RIT* integrates edge costs along the straight line $\bar{\sigma}_{uv}$ rather than the true geodesic, yielding $\bar{c}(u, v) \geq d_R(u, v)$. Rewiring (9) and pruning (10) use $\bar{c}$. *The heuristic is implemented as follows: for constant metrics (Euclidean or diagonal anisotropic), $\hat{h}_R(v) = \bar{c}(v, x_g)$ (the L_1 midpoint estimate, which is exact); for spatially-varying metrics, the implementation uses the admissible lower bound $\hat{h}_R(v) = \sqrt{\lambda_{\min}} \|v - x_g\|_2$, where $\lambda_{\min}$ is the smallest eigenvalue of G over the configuration space. This lower bound is admissible (never overestimates the true cost-to-go), ensuring the heuristic does not incorrectly deprioritise vertices with high actual cost-to-come. Since $\bar{c} \geq d_R$, the $\bar{c}$ -evaluated pruning rule is not pointwise equivalent to (10); it is asymptotically consistent, however, because by Lemma 1 the per-edge bias is $O(\|\nabla G\|_\infty \|v - u\|^2)$ and $\bar{c} \rightarrow d_R$ along the optimal path as edges shorten, so pruning error vanishes asymptotically as edge lengths decrease. Asymptotic optimality is preserved (Theorem 3). The bias is negligible for the short edges (≤ 0.1 rad) used in our manipulation benchmarks.*

D. Tree Rewiring and Pruning

Rewiring (line 11 of algorithm 1): When a new vertex v is added to the tree, it may provide a cheaper path to existing nearby vertices. For each neighbour u within the Riemannian radius r_n^R , RIT* checks whether the path through v is cheaper than the current parent of u :

$$g(v) + c_R(v, u) < g(u), \quad (8)$$

where $g(\cdot)$ is the Riemannian cost-to-come from x_s along the current tree and $c_R(v, u)$ is the Riemannian edge cost (1) between v and u . If this condition holds, u is reconnected as a child of v , and the cost-to-come of all descendants is updated. Because this comparison uses c_R rather than Euclidean distance, rewiring respects the anisotropic cost structure and discovers connections that Euclidean planners would overlook.

Pruning (line 5 of algorithm 1): After each batch, RIT* removes vertices that can no longer contribute to an improved solution. A vertex x is pruned if its best possible cost-to-come plus cost-to-go exceeds the current best solution cost:

$$d_R(x_s, x) + d_R(x, x_g) > c_{\text{best}}. \quad (9)$$

Equation (10) is the idealised criterion; **in implementation the stored f -value $f(x) = g(x) + \hat{h}_R(x)$ is used as a surrogate for $d_R(x_s, x) + d_R(x, x_g)$, and pruning applies a soft threshold: a vertex is removed when $f(x) > \tau \cdot c_{\text{best}}$, where $\tau = 1.05$ is a fixed slack factor. This slightly softer rule retains near-boundary vertices across batches and is asymptotically consistent with (10) as $f(x) \rightarrow d_R(x_s, x) + d_R(x, x_g)$ under densification (Lemma 1), preserving asymptotic optimality (Theorem 3).** That is, a vertex is removed when it falls outside the current Riemannian informed set I_R . Pruning reduces the tree size and nearest-neighbour search cost, and when CARM updates the metric (line 20 of algorithm 1), it forces re-evaluation of vertex membership under the tightened I_R , ensuring the tree adapts to the evolving cost landscape.

E. Collision-Adaptive Riemannian Metric (CARM)

The preceding components assume that the metric G is known a priori. When it is not, RIT* learns one online via CARM (algorithm 2), which takes four inputs: a set of collision points $\mathcal{C}_{\text{col}} = \{c_i\}_{i=1}^N$ collected from failed edge checks, the current metric G , a kernel bandwidth σ that controls spatial resolution, and an inflation strength α that controls how aggressively traversal cost increases near obstacles.

The key observation is that failed edge checks naturally reveal points near obstacle boundaries. As these accumulate, they form a sparse map of regions where traversal should be expensive. CARM converts this map into a density field using a Gaussian kernel density estimate (KDE),

$$\hat{f}(x) = \frac{1}{N} \sum_{i=1}^N \exp\left(-\frac{\|x - c_i\|^2}{2\sigma^2}\right), \quad (10)$$

and applies it as a conformal scaling of the base metric (lines 1-3 of algorithm 2):

$$G_{\text{CARM}}(x) = s(x) G(x), \quad s(x) = 1 + \alpha \hat{f}(x), \quad (11)$$

figures/pruning.png

Fig. 3: Tree rewiring and pruning in RIT*. (a) Rewiring updates parent connections when a lower-cost Riemannian path is found. (b) Pruning removes vertices outside the Riemannian informed set, reducing tree size and focusing the search.

Algorithm 2: CARM_UPDATE($\mathcal{C}_{\text{col}}, G, \sigma, \alpha$)

Input: Collision points $\mathcal{C}_{\text{col}} = \{c_i\}_{i=1}^N$, current metric G , bandwidth σ , strength α
Output: Updated metric G_{CARM}
1 Build KDE: $\hat{f}(x) \leftarrow N^{-1} \sum_{i=1}^N \exp(-\|x - c_i\|^2 / (2\sigma^2))$
2 Compute scale field: $s(x) \leftarrow 1 + \alpha \hat{f}(x)$
3 Update metric: $G_{\text{CARM}}(x) \leftarrow s(x) G(x)$
4 Recompute metric cache at grid points
5 Recompute I_R with G_{CARM}
6 $G \leftarrow \text{UpdateG}(G_{\text{CARM}})$
7 **return** G

where $G(x)$ defaults to I_d if no prior metric is available. Because $s(x)$ is scalar, G_{CARM} is a *conformal* refinement: CARM learns an obstacle-proximity inflation field, not anisotropic tensor structure. Directional cost variation, when present, must come from the base metric $G(x)$; this is appropriate for collision-based learning because collisions carry positional but not directional information. In algorithm 2, G_{CARM} denotes the updated metric locally; once returned, it overwrites G in the main loop (line 20 of algorithm 1), so all subsequent sampling, edge evaluation, and pruning use the CARM-refined metric seamlessly. The updated metric satisfies $G_{\text{CARM}}(x) \succeq G(x)$, which increases traversal cost relative to the base metric and therefore contracts the informed region under the refined metric. Completeness and asymptotic optimality are therefore preserved (Corollary 4).

After updating the metric, the algorithm recomputes the metric cache, shrinks I_R under the new cost field, and prunes vertices that fall outside the tightened informed set (lines 4-6 of algorithm 2). This creates a feedback loop: collisions refine the metric, the refined metric focuses sampling, and focused sampling generates more informative collisions.

CARM is triggered every K_c iterations (line 20 of algorithm 1) and only when the collision buffer is non-empty.

TABLE I: Experimental environment summary. d : dimension; κ : metric condition number; ℓ_{max} : max edge length; n_c : collision checks per edge.

Environment	d	κ	ℓ_{max}	n_c
2-D Random World	2	1	0.05	10
3-D Diagonal	3	≈ 6	0.05	10
6-D UR10	6	≈ 12	0.10 rad	10
14-D Tiago Bimanual	14	≈ 18	0.10 rad	10

As fig. 4 shows, the KDE field stabilises after approximately 20 iterations (~ 400 collision points in 2-D), after which, further updates produce negligible change. The stopping criterion $\|\hat{f}_t - \hat{f}_{t-1}\|_\infty < \varepsilon_0$ ($\varepsilon_0 = 10^{-3}$) is met within 20 ± 3 updates across all benchmarks in our experiments. We use $\sigma = 0.1$, $\alpha = 10$, and $K_c = 15$ across all environments; section V-E reports sensitivity to these choices.

F. Theoretical Analysis

RIT* preserves the asymptotic guarantees of BIT* [4]; we sketch the modifications needed under a Riemannian metric and CARM.

Assumption 1. G satisfies $\lambda_{\min} I_d \preceq G(x) \preceq \lambda_{\max} I_d$ on $\mathcal{C}_{\text{free}}$ with $0 < \lambda_{\min} \leq \lambda_{\max} < \infty$ and $\|\nabla G\|_\infty < \infty$; the optimal path σ^* has cost c^* and strong δ -clearance. For inertia metrics $G(q) = M(q)$ this holds when the configuration bounds exclude kinematic singularities; reported $\kappa = \lambda_{\max}/\lambda_{\min}$ is the maximum over 10^4 grid samples.

Bounding (1) by the extreme eigenvalues gives the bi-Lipschitz equivalence.

$$\sqrt{\lambda_{\min}} \|x - y\|_2 \leq d_R(x, y) \leq \sqrt{\lambda_{\max}} \|x - y\|_2, \quad (12)$$

Lemma 1 (Edge bias of the straight-line approximation). *For any $u, v \in \mathcal{C}_{\text{free}}$,*

$$0 \leq \bar{c}(u, v) - d_R(u, v) \leq \frac{1}{2} \sqrt{\lambda_{\max}} \|\nabla G\|_\infty \|v - u\|^2. \quad (13)$$

The lower bound is immediate; the upper bound follows from a second-order expansion of (1) around the geodesic with deviation $\|\bar{\sigma}_{uv} - \gamma_{uv}^\| = O(\|v - u\|)$ (where γ_{uv}^* denotes the u -to- v geodesic) under bounded curvature. Hence $\bar{c} \rightarrow d_R$ as $\|v - u\| \rightarrow 0$, and the $\bar{c}$ -evaluated pruning and heuristic converge to their d_R counterparts along σ^* under densification, so no vertex of σ^* is permanently pruned.*

Theorem 2 (Probabilistic completeness). *RIT* finds a feasible solution with probability 1 as $q \rightarrow \infty$.*

Proof. Discretize σ^* into waypoints $w_0, \dots, w_K$ with $\|w_k - w_{k+1}\|_2 < \delta/2$; each is in the fixed set $I_R^+ = \{x : \bar{c}(x_s, x) + \bar{c}(x, x_g) \leq c^* + \varepsilon\}$, which has a positive measure and is contained in I_R . Uniform sampling over I_R hits every $\delta/4$ -ball almost surely; by (13) the nearest samples satisfy $d_R(s_k, s_{k+1}) < \sqrt{\lambda_{\max}} \delta$, which falls below the connection radius $r_n^R = \Theta((\log q/q)^{1/d})$ for sufficiently large q , yielding a collision-free chain $s_0 \rightarrow \dots \rightarrow s_K$ almost surely. $\square$

Theorem 3 (Asymptotic optimality). $P(\lim_{q \rightarrow \infty} c_q^{\text{RIT}^*} = c^*) = 1$.

TABLE II: Benchmark Evaluations. Per environment: $t_{\text{init}}^{\min}$, $t_{\text{init}}^{\text{med}}$ (min/median time to first solution, s), $c_{\text{init}}^{\text{med}}$ (median initial cost), and $c_{\text{fin}}^{\text{med}}$ (median final cost). **Bold**: best per column.

Planner	2-D Random World ($\mathbb{R}^2$)				3-D Diagonal ($\mathbb{R}^3$)				6-D UR10 ($\mathbb{R}^6$)				14-D Tiago Biman. ($\mathbb{R}^{14}$)				Succ. (%)
	$t_{\min}$	t_{med}	$c_{\text{init}}^{\text{med}}$	$c_{\text{fin}}^{\text{med}}$	$t_{\min}$	t_{med}	$c_{\text{init}}^{\text{med}}$	$c_{\text{fin}}^{\text{med}}$	$t_{\min}$	t_{med}	$c_{\text{init}}^{\text{med}}$	$c_{\text{fin}}^{\text{med}}$	$t_{\min}$	t_{med}	$c_{\text{init}}^{\text{med}}$	$c_{\text{fin}}^{\text{med}}$	
RIT*	0.013	0.020	0.9578	0.9135	0.021	0.040	2.4923	2.1608	0.0601	0.0915	1.9143	1.4143	0.8135	3.4255	16.5747	8.5778	100
BIT*	0.0183	0.0308	0.9698	0.9108	0.0476	0.0698	2.8167	2.2027	0.0651	0.0820	1.9415	1.5415	0.7657	0.8834	18.5069	10.7092	100
Inf. RRT*	0.0129	0.0289	0.9946	0.9165	0.0312	0.0464	2.8263	2.2278	0.1390	0.2114	1.9541	1.6975	1.5915	2.2198	19.9509	13.5773	100
AIT*	0.0111	0.0217	1.0475	0.9116	0.0236	0.0433	2.9950	2.2563	0.0697	0.1159	2.7794	1.4588	0.8070	1.4469	18.8514	11.0986	100
EIT*	0.0106	0.0162	1.0285	0.9124	0.0247	0.0413	2.8736	2.2449	0.0705	0.1397	2.7929	1.4530	0.8095	1.4452	19.8802	11.4415	100
APT*	0.0127	0.0189	1.0946	0.9225	0.0312	0.0465	2.7233	2.4436	0.1391	0.2126	2.9256	1.7536	1.6003	2.2284	19.8939	14.0232	100

figures/fig_carm_evolution.png

Fig. 4: CARM metric evolution over planning iterations. As collision points accumulate, the learned cost field progressively refines, tightening the informed set and improving sample allocation.

Proof. We verify the Karaman-Frazzoli conditions [2]. By (13), a Riemannian ball of radius r_n^R contains a Euclidean ball of radius $r_n^R/\sqrt{\lambda_{\max}}$, so RIT*'s edge set is a superset of BIT*'s at $\bar{r}_q = r_n^R/\sqrt{\lambda_{\max}}$ and inherits its neighbourhood properties. Using (6), $\mu_R(I_R) \geq \lambda_{\min}^{d/2} \mu(I_R) > 0$, so the connection-radius constant exceeds the Karaman-Frazzoli threshold. The remaining steps (dense approximation of σ^* , cost convergence, optimal rewiring) follow [4] with d_R replacing $\|\cdot\|_2$ and (13) together with Lemma 1 bounding the approximation error. $\square$

Corollary 4 (CARM preserves guarantees). *With CARM, $1 \leq s(x) \leq 1 + \alpha$, so the scaled metric satisfies Assumption 1 with $(\lambda_{\min}, (1 + \alpha)\lambda_{\max})$. Updates cease at iteration t_0 when $\|\hat{f}_t - \hat{f}_{t-1}\|_\infty < \varepsilon_0$; since at most B collision points enter per K_c -window, the per-update change is $O(B/N_t) \rightarrow 0$, so $t_0 < \infty$ almost surely (trivially if no further collisions occur). For $t \geq t_0$, the metric is frozen at G_∞ and Theorems 2 and 3 apply with its constants; Since $G_\infty \succeq G$, the refined metric induces no-smaller traversal costs than the base metric, so post-CARM pruning never removes configurations on improving paths.*

V. EXPERIMENTS

A. Experimental Setup

We evaluated RIT* in environments that span 2-D, 3-D, 6-D and 14-D configuration spaces (Table I), designed to assess both the Riemannian informed-set construction and the CARM adaptation mechanism¹.

In low-dimensional spaces ($\mathbb{R}^2$ and $\mathbb{R}^3$), we consider two geometric benchmarks. The *Random World* (2-D) consists of a square populated with 30 obstacles under a Euclidean metric ($\kappa = 1$), providing an isotropic baseline. The *Diagonal* environment (3-D) is a unit cube with obstacles aligned with the axis that form a narrow passage, equipped with a diagonal-anisotropic metric $G = \text{diag}(1, 3, 6)$ ($\kappa \approx 6$) to evaluate direction-dependent costs.

In higher-dimensional manipulation tasks ($\mathbb{R}^6$ and $\mathbb{R}^{14}$), we evaluate scalability under articulated robot dynamics. In $\mathbb{R}^6$, we use a UR10e arm with a Robotiq 85 gripper in PyBullet [16], with a joint-space inertia metric $G(q) = M(q)$ ($\kappa \approx 12$), maximum edge length 0.1 rad, and 10 collision checks per edge. The 6-D task (*UR10 Drill*, Table II) retrieves a drill from a cluttered shelf.

In $\mathbb{R}^{14}$, we used a *Tiago Pro Bimanual* task with two 7-DOF arms performing a coordinated pre-grasp motion toward a tall box, with obstacle pillars blocking direct paths. The combined joint space uses a bimanual inertia metric ($\kappa \approx 18$), representing our most challenging setting in terms of dimensionality and anisotropy.

We compare RIT* with five asymptotically optimal anytime planners: Informed RRT* [3], BIT* [4], AIT*, EIT* [5], and APT* [6]. All planners are implemented in a common Python framework with shared collision checks, random seeds, and environment interfaces for fair comparison. Each method is provided with the same metric tensor G , and all planners evaluate edge costs using the same Riemannian integral (1), ensuring a common objective. The cascading evaluation of Sec. IV-C is applied to all planners; reported time differences are therefore attributable to informed-set geometry and connectivity, not to integration cost. We intentionally retain the original Euclidean informed-set formulations of the baselines to isolate the effect of replacing these components with Riemannian counterparts within an otherwise unchanged planning pipeline. All planners use batch size $B = 100$, maximum iterations $T = 200$, a per-trial timeout of 200 s, and base seed 60. The edge parameters follow Table I. Baselines differ only in their use of Euclidean informed sets and nearest-neighbour radii.

¹Code and scripts to reproduce all reported results are provided as supplementary material accompanying this submission. The implementation will be released publicly upon acceptance

figures/environments.png

Fig. 5: Benchmark environments from 2-D to 14-D, covering geometric setups and manipulation tasks for RIT* evaluation.

For RIT*, we use a diagonal approximation of the metric, where $G(q)$ is evaluated at grid points via trilinear interpolation from a precomputed cache. This avoids expensive geodesic computation and reduces the cost of repeated metric evaluations, which is effective for smoothly varying metrics (e.g., diagonal and inertia metrics). The cache resolution is set to $\text{RES}=32$ (2-D/3-D) and $\text{RES}=3$ (6-D/14-D). The coarse 6-D/14-D resolution is justified by the smooth variation of $M(q)$ within the planning bounds (relative interpolation error $<4\%$ at 10^4 random configurations), and because the cache feeds only the L_1 filter; L_3 re-samples $G(q)$ on the fly.

Each experiment uses $N = 50$ trials with a 200 s timeout. We report wall-clock time and Riemannian path cost; *init/final* denote the first and terminal solutions. The **bold** indicates the best per column among planners. The metric cache stores $G(x)$ on a res^d grid (resolution res per dimension) with $O(1)$ lookup. The connection radius is computed using the Riemannian volume approximation $\mu_R(I_R) \approx \det(\bar{G})^{1/2} \mu(I_E)$. This approximation is exact for spatially constant G and incurs error $O(\|\nabla G\|_\infty \text{diam}(I_R))$ otherwise; on our benchmarks (spatially-uniform diagonal/inertia metrics within bounds) the error is $<1\%$, and on the controlled β -sweep of Sec. IV-A it reaches 2.7% at $\beta = 1$, consistent with the bound.

B. Low-Dimensional Benchmarks (2-D, 3-D)

In isotropic 2-D ($\kappa = 1$), RIT* attains the lowest cost (0.958), with gaps to baselines ranging 1.3-14.3% (BIT* +1.3%, IRRT* +3.8%, EIT* +7.4%, AIT* +9.4%, APT* +14.3%); Euclidean and Riemannian informed sets coincide here, so the small lead reflects only the tighter NN ellipsoid. At convergence, all planners cluster within 1.3% ($c_{\text{fin}}^{\text{med}} \in [0.911, 0.923]$), with BIT* marginally lowest (0.911 vs. RIT* 0.914), confirming the predicted equivalence under isotropy. In 3-D ($\kappa \approx 6$), RIT* attains the lowest cost (2.49) and is

TABLE III: Ablation study: mean path cost ($\pm$ std) over 50 trials. **Bold**: best practical variant with complete success (oracle excluded). “w/o CARM” uses the oracle metric.

Variant	2-D	3-D	6-D	14-D
RIT* (full)	0.921 $\pm$ 0.007	2.196 $\pm$ 0.027	1.489 $\pm$ 0.082	8.638 $\pm$ 0.144
w/o Riem. samp.	0.916 $\pm$ 0.002	2.388 $\pm$ 0.182	1.459 $\pm$ 0.075	9.226 $\pm$ 0.935
w/o cascading	0.920 $\pm$ 0.008	2.195 $\pm$ 0.017	1.489 $\pm$ 0.082	8.638 $\pm$ 0.144
w/o CARM	0.913 $\pm$ 0.003	2.174 $\pm$ 0.033	1.484 $\pm$ 0.086	8.586 $\pm$ 0.012

fastest to a first solution (0.040 s), outperforming APT* (2.72, +9.3%), BIT* (2.82, +13.0%), IRRT* (2.83, +13.4%), EIT* (2.87, +15.3%), and AIT* (2.99, +20.1%). The 3-D Diagonal metric is spatially constant, so the mean-metric whitening of Sec. IV-A is exact and the Riemannian informed set coincides with a uniformly rescaled I_E ; the gain over BIT* in this regime is attributable to RIT*’s tighter nearest-neighbour ellipsoid rather than informed-set tightening, which becomes dominant when the metric varies spatially (Sec. V-C). The lead persists at convergence: RIT* attains the lowest $c_{\text{fin}}^{\text{med}} = 2.16$, with gaps of +1.9% (BIT*), +3.1% (IRRT*), +3.9% (EIT*), +4.4% (AIT*), and +13.1% (APT*).

C. High-Dimensional Benchmarks (6-D, 14-D)

In the 6-D UR10 task ($\kappa \approx 12$), RIT* achieves the lowest cost across all baselines (1.91) while maintaining complete success (100%); RIT* improves on BIT* (1.94, +1.4%) and IRRT* (1.95, +2.1%); AIT* (2.78, +45.2%), EIT* (2.79, +45.9%), and APT* (2.93, +52.8%) trail further. At convergence RIT* retains the lead ($c_{\text{fin}}^{\text{med}}$: RIT* 1.414, EIT* 1.453, AIT* 1.459; spread 3.2%), with BIT* (+9.0%), IRRT* (+20.0%), and APT* (+24.0%) trailing; the Riemannian informed set tightens both early progress and final convergence in this regime. In the 14-D Tiago task ($\kappa \approx 18$), RIT* attains the lowest cost (8.58), outperforming BIT* (10.71, +24.8%), AIT* (11.10, +29.4%), and EIT* (11.44, +33.4%), with IRRT* (13.58, +58.3%) and APT* (14.02, +63.5%) trailing further. The same ordering holds for $c_{\text{fin}}^{\text{med}}$, where RIT*’s lead remains the largest across all environments, indicating that informed-set tightening, not just early connectivity, drives the gain at high dimension and anisotropy.

Performance gains grow with both anisotropy and dimensionality: up to +14.3% (2-D), +20.1% (3-D), up to +24.0% in 6-D, and +24.8-63.5% (14-D, across all five baselines).

Real robot execution: We additionally demonstrate execution of RIT*-planned trajectories on a UR10e platform (Fig. 7); a full hardware comparison against baselines is left to future work. The executed trajectories are smooth, collision-free, and preferentially utilize lower-inertia joints (wrist over shoulder/elbow), consistent with simulation results. Across 20 hardware runs of three planned trajectories, all were collision-free, with mean joint tracking error 0.018 ± 0.004 rad and end-effector deviation 4.2 ± 1.1 mm.

D. Ablation Study

We isolate the contribution of each component across representative 2-D, 3-D, 6-D, and 14-D environments (Table III): (i) RIT* (full), (ii) w/o Riemannian sampling (Euclidean informed

figures/cost.png

Fig. 6: Convergence and success rate across environments. Top: success rate vs. time; bottom: Riemannian path cost vs. time (median, spread over trials). RIT* converges faster and to lower-cost solutions, with the largest gains at high dimension and anisotropy (6-D, 14-D).

set), (iii) *w/o cascading* (full edge evaluation), and (iv) *w/o CARM* (oracle metric). The oracle variant uses the ground-truth metric and serves as an upper bound rather than a directly comparable method.

Observations: Low-D: Riemannian sampling provides the dominant benefit under anisotropy (+8.7% in 3-D without it; negligible in isotropic 2-D, where removing it is marginally lower (0.916 vs. 0.921)). CARM closely tracks the oracle in every environment (< 1.1% overhead), with the largest residuals in 2-D and 3-D (+0.9%, +1.0%), where the cost

field is learned from $G = I_d$. *High-D:* gains are smaller, and performance is primarily governed by informed-set geometry; CARM provides incremental refinement over the base metric. *Cascading* has < 0.3% impact on cost, confirming it is primarily a computational optimisation. These trends are consistent with the analysis in Sec. V-E.

E. CARM Analysis

We evaluate how well CARM approximates the oracle (ground-truth) metric by comparing RIT* under learned ver-

figures/real-ur10.png

Fig. 7: UR10e real-robot execution of RIT*-planned trajectories. The planner produces smooth, collision-free, anisotropy-aware motions that transfer directly from simulation to hardware.

TABLE IV: CARM quality analysis: mean path cost (oracle vs. learned metric). Overhead denotes relative cost increase w.r.t. the oracle; lower is better.

Env	Oracle	CARM	Overhead
2-D	0.913 ± 0.003	0.921 ± 0.007	+0.9%
3-D	2.174 ± 0.033	2.196 ± 0.027	+1.0%
6-D	1.484 ± 0.086	1.489 ± 0.082	+0.3%
14-D	8.586 ± 0.012	8.638 ± 0.144	+0.6%

sus oracle metrics (Table IV). The interpretation differs by dimension. In 6-D/14-D, the oracle is the inertia matrix $M(q)$ and CARM applies a learned proximity-based scaling to this same $M(q)$, so the small overhead ($< 1\%$) reflects the near-optimality of the base metric rather than CARM recovering the full metric structure a stricter test would use $G = I_d$.

The low-dimensional case is therefore the most informative. Starting from $G = I_d$ in 2-D, CARM must reconstruct the obstacle-proximity cost entirely from collision feedback, yet attains overhead of only +0.9%, and +1.0% in 3-D. The 6-D/14-D gap ($< 1\%$) then represents incremental refinement over an already strong prior. Across all four settings CARM stays within $\sim 1\%$ of the oracle, indicating that the learned scalar field recovers nearly all the obstacle-proximity information available from collisions, even when starting from scratch. In all settings, CARM learns a scalar cost field that modulates a base metric rather than a fully anisotropic tensor an appropriate choice, since collisions carry positional but not directional information.

Two additional checks support these conclusions, with $\sigma = 0.1$, $\alpha = 10$, $K_c = 15$ held fixed throughout: (a) a

$3 \times 3 \times 3$ sweep over $\sigma \in \{0.05, 0.1, 0.2\}$, $\alpha \in \{5, 10, 20\}$, $K_c \in \{10, 15, 30\}$ on the 3-D Diagonal benchmark varies median cost by less than $\pm 2\%$, indicating low hyperparameter sensitivity; (b) re-running 6-D UR10 with $G = I_d$ (no inertia prior), RIT*+CARM attains median cost 1.451 vs. 1.414 for the oracle (+2.6%) and 1.520 without CARM (+7.5%), so CARM recovers $\sim 2/3$ of the oracle gap from collisions alone.

VI. DISCUSSION AND LIMITATIONS

A. Discussion

The benefit of RIT* increases with both anisotropy (κ) and dimensionality. For isotropic costs ($\kappa \approx 1$), the Riemannian and Euclidean informed sets coincide ($I_R \approx I_E$), and the gap to informed-set baselines is small (within 7.4%) although weaker informed-sampling strategies (APT*) trail by up to 14.3%. As anisotropy increases, the Riemannian informed set becomes progressively better aligned with the underlying cost structure, concentrating samples along cost-relevant directions: RIT* achieves +13.0% over BIT* in 3-D ($\kappa \approx 6$). In high-dimensional settings (14-D, $\kappa \approx 18$) the advantage is largest: RIT* attains +24.8-63.5% over all five baselines, with the spread widening as Euclidean-informed planners lag the cost geometry. Overall, the gain scales with the tightening of the informed set under the underlying cost geometry. RIT* also maintains competitive convergence speed, as shown in Fig. 6, despite the additional metric computations.

B. Limitations

Some limitations suggest directions for improvement. (i) The metric cache scales as res^d , which can become prohibitive in high dimensions; sparse, adaptive, or learned representations could mitigate this. (ii) CARM employs isotropic Gaussian kernels, which may blur directional structure near obstacles; anisotropic kernels aligned with local geometry could improve fidelity. (iii) CARM relies on collision feedback, and its effectiveness depends on the informativeness of sampled collisions; in simpler or low-dimensional settings, recovery may be slower, motivating exploration strategies that actively probe uncertain regions. (iv) The computational overhead of metric evaluation and caching may impact scalability in very high dimensions.

VII. CONCLUSIONS

We introduced RIT*, a Riemannian batch-informed planner that replaces Euclidean primitives with cost-consistent counterparts and learns a collision-adaptive conformal cost refinement online via CARM. Across environments from 2-D to 14-D, RIT* consistently produces lower-cost solutions under anisotropic metrics while remaining competitive in isotropic settings, with gains scaling with anisotropy and dimension; +13.0% over BIT* in 3-D, up to +24.0% in 6-D manipulation, and +24.8-63.5% in 14-D bimanual planning, where Euclidean-informed baselines trail substantially. CARM recovers a substantial portion of the oracle-metric advantage from collision feedback alone, enabling cost-aware planning without prior knowledge of the cost field.

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